

State of GPT ｜ BRK216HFS

State of GPT（字幕版）｜BRK216HFS：从预训练到 RLHF，大语言模型的全景解读

时长 **41:34** 句数 **1042** 术语 **20** 章节 **16**

摘要

本讲共 16 章，从「神经网络 · 训练集」到「生成式预训练Transformer模型」。全文 1042 句、1042 句已译成中文，标注了 20 个 AI 术语（主要集中在神经网络基础、数学与微积分、训练与优化）。

16 chapters, 1042 sentences (1042 translated), 20 AI terms annotated. Built fully offline with opus-mt + local glossary.

神经网络 · 训练集 00:07

00:07 Please welcome AI researcher and founding member of OpenAI,
欢迎AI研究员和OpenAI创始成员

00:12 Andre Carpathi.
安德烈·卡巴提 安德列·卡帕蒂

00:23 I'm happy to be here to tell you about the state of GPT
很高兴能在此 告诉你们GPT的状态

> **gpt**
> 生成式预训练Transformer模型 生成式预训练Transformer模型
> 神经网络基础

00:27 and more generally about the rapidly growing ecosystem
更广义地说,关于迅速增长的生态系统

00:30 of large language models.
大型语言模型。

00:32 So I would like to partition the talk into two parts.
因此,我想把谈话分成两部分。

00:35 In the first part, I would like to tell you
在第一部分,我想告诉你

00:36 about how we train GPT assistance.
如何训练GPT援助。

gpt
生成式预训练Transformer模型 生成式预训练Transformer模型
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00:39 And then in the second part, we
然后在第二部分,我们

00:41 are going to take a look at how we can use these assistance
将研究我们如何利用这些援助

00:43 effectively for your applications.
有效申请。

00:46 So first, let's take a look at the emerging recipe
首先,让我们先看看 新兴的食谱

00:49 for how to train these assistance.
如何培训这些援助。

00:50 And keep in mind that this is all very new and still
记住,这一切都是全新的,仍然

00:52 rapidly evolving.
迅速演变。

00:53 But so far, the recipe looks something like this.
但到目前为止,食谱看起来是这样的

00:55 Now, this is kind of a complicated slide,
现在,这是一个复杂的幻灯片

00:57 so I'm going to go through it piece by piece.
所以我要逐个地通过它

01:00 But roughly speaking, we have four major stages,
但大致而言,我们有四个主要阶段:

01:03 pre- training , supervised fine-tuning, reward modeling,

预训前,监督的微调,奖励模型,

training ['treiniŋ]

训练 确定构成模型的理想参数 的过程。

神经网络基础

01:06 reinforcement learning , and they follow each other

学习加强,互相跟随

reinforcement learning

强化学习 强化学习

神经网络基础

01:08 serially.

连续的

01:10 Now, in each stage, we have a data set that

现在,在每一个阶段,我们都有一套数据

data set

数据集 一组样本 的集合。 Dataset API (tf.data) 一种高级别的 TensorFlow API ，用于读取数据并将其转换为机器学习算法所需的格式。 tf.data.Dataset 对象表示一系列元素，其中每个元素都包含一个或多个张量 。 tf.data.Iterator 对象可获取 Dataset 中的元素。 如需详细了解 D

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01:13 powers that stage.

使这个阶段具有权力。

01:14 We have an algorithm that for our purposes

我们有一个算法,为了我们的目的

algorithm ['ælgəriðm]

算法 An effective procedure: a finite sequence of unambiguous steps that, when carried out on a given input, produces the intended output and terminates. The word derives from

神经网络基础

01:16 will be a objective for training a neural network.
将成为培训神经网络的目标。

training ['treiniŋ]
训练 确定构成模型的理想参数 的过程。
神经网络基础

neural network
神经网络 一种模型，灵感来源于脑部结构，由多个层构成（至少有一个是隐藏层），每 个层都包含简单相连的单元或神经元 （具有非线性关系）。
神经网络基础

01:22 And then we have a resulting model.
然后我们有了一个结果模型。

01:24 And then there are some nodes on the bottom.
底部还有一些节点

01:25 So the first stage we're going to start with
所以第一个阶段 我们从

01:27 is the pre- training stage.
培训前阶段。

training ['treiniŋ]
训练 确定构成模型的理想参数 的过程。
神经网络基础

01:28 Now, this stage is kind of special in this diagram.
现在,这个阶段在这个图表中 是一种特殊的阶段。

01:31 And this diagram is not to scale,
这个图表不是用来缩放的,

01:33 because this stage is where all of the computational work
因为这个阶段是所有计算工作都在这里

01:35 basically happens.
基本上就是这样

01:36 This is 99% of the training compute time and also flops.
这是训练计算时间的99% 也是Flops

training ['treiniŋ]
训练 确定构成模型的理想参数 的过程。
神经网络基础

01:41 And so this is where we are dealing with internet-scale data
所以我们就是在这里处理 互联网数据

01:45 sets with thousands of GPUs in the supercomputer
超级计算机中设置了数千个 GPU

01:48 and also months of training, potentially.
可能还有几个月的培训时间。

training ['treiniŋ]
训练 确定构成模型的理想参数 的过程。
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01:51 The other three stages are fine-tuning stages
其他三个阶段是微调阶段

01:53 that are much more along the lines of small few numbers
少数国家的情况,

01:56 of GPUs and hours or days.
和小时或天数。

01:59 So let's take a look at the pre-training stage
让我们看看训练前阶段

training ['treiniŋ]
训练 确定构成模型的理想参数 的过程。
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02:00 to achieve a base model.
实现一个基准模式。

02:04 First, we're going to gather a large amount of data.
首先,我们将收集大量的数据。

02:07 Here's an example of what we call a data mixture that
下面的例子 我们称之为数据混合

02:10 comes from this paper that was released by Meta, where they
本文来自梅塔发布的论文,他们在那里

02:14 released this Lama-based model.
释放了这个基于喇嘛的模式。

02:16 Now, you can see roughly the kinds of data sets
现在,你可以大致看到 数据集的类型

02:18 that enter into these collections.
输入这些收藏。

02:20 So we have common crawl, which is just a web scrape, C4, which
所以我们有共同的爬行, 这只是一个网刮,C4,

02:23 is also common crawl, and then some high-quality data sets
也是常见的爬行,然后是一些高质量的数据集

02:26 as well.
也好,我也会

02:27 So for example, GitHub, Wikipedia, Books, Archive,
例如,吉特胡布、维基百科、书籍、档案馆等。

02:30 Stack Exchange, and so on.
堆叠交换,等等。

02:31 These are all mixed up together.
这些都是混在一起的

02:33 And then they are sampled according
然后按照

02:34 to some given proportions.
一定比例的计算。

02:36 And that forms the training set for the neural net for the GPT.
形成GPT神经网的训练设置

training ['treiniŋ]
训练 确定构成模型的理想参数 的过程。
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neural net
神经网络 神经网络
神经网络基础

02:41 Now, before we can actually train on this data,
现在,在我们真正能训练数据之前

02:43 we need to go through one more pre-processing step,
我们需要再经过一个预处理步骤

02:45 and that is tokenization.
那是象征意义。

02:46 And this is basically a translation of the raw text
这基本上是原始文本的翻译

02:49 that we scrape from the internet into sequences of integers,
我们从互联网上刮到的,是整数的序列,

02:53 because that's the native representation over which GPTs
因为这就是当地代表 通用计算机

02:56 function.
函数。

02:57 Now, this is a lossless kind of translation
现在,这是无亏无亏的翻译

03:00 between pieces of text and tokens and integers.
在文本和图象以及整数之间。

03:03 And there are a number of algorithms for this stage,
这个阶段有许多算法

03:05 typically, for example, you could use something
通常,例如,你可以用一些东西

03:07 like byte-period encoding, which iteratively merges
象字节期编码,它迭接合并

03:10 little text chunks and groups them into tokens.
将小块文字分组成符号。

03:13 And so here I'm showing some example chunks of these tokens,
所以我在这里展示一些例子 部分这些象征物,

03:16 and then this is the raw integer sequence that will actually
然后,这是 原始整数序列,它实际上将会

03:19 feed into a transformer.
输入变压器。

transformer [træns'fɔ:mə]
Transformer 中文为"变换器"，但不够达雅，日常还是称英文 Transformer 中文为"变换器"，但不够达雅，日常还是称英文
神经网络基础

03:22 Now, here I'm showing two examples
现在,我在这里举两个例子

03:25 for hyper-primers that govern this stage.
给主导本阶段的超优等豪华者

训练 03:28

03:28 So GPT-4, we did not release too much information about how
GPT-4,我们没有透露太多关于

gpt
生成式预训练Transformer模型 生成式预训练Transformer模型
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03:31 it was trained, and so on.
训练过,等等

03:32 So I'm using GPT-3's numbers,
所以我用GPT -3的号码

gpt
生成式预训练Transformer模型 生成式预训练Transformer模型
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03:33 but GPT-3 is, of course, a little bit old by now,
但GPT-3现在当然有点老了

gpt
生成式预训练Transformer模型 生成式预训练Transformer模型
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03:36 about three years ago.
大约三年前

03:37 But Lama is a fairly recent model from Meta.
但拉马是梅塔最近的一个模型

03:40 So these are roughly the orders of magnitude
所以这些大致是数量级

03:42 that we're dealing with when we're doing pre- training .
训练前我们面对的就是训练前

training ['treiniŋ]
训练 确定构成模型的理想参数 的过程。
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03:45 The vocabulary size is usually a couple 10,000 tokens.
词汇大小通常为一万个符号。

03:48 The context length is usually something like 2,000, 4,000,
上下文长度通常大约2000,4,000

03:51 or nowadays even 100,000.
或现在甚至有十万

03:53 And this governs the maximum number of integers
以此来调节最大整数

03:56 that the GPT will look at when
通用一般观察点将看到

gpt
生成式预训练Transformer模型 生成式预训练Transformer模型
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03:58 it's trying to predict the next integer in a sequence.
它试图按顺序预测下一个整数。

04:02 You can see that roughly the number of parameters
您可以看到参数的大致数量

04:04 is, say, 65 billion for Lama.
也就是说,拉马有650亿

04:06 Now, even though Lama has only 65B parameters compared
现在,尽管喇嘛的参数只有65B

04:08 to GPT-3's 175 billion parameters,
GPT-3的7500亿参数

gpt
生成式预训练Transformer模型 生成式预训练Transformer模型
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04:11 Lama is a significantly more powerful model.
拉马是一个更强大的模型。

04:14 And intuitively, that's because the model
直觉上,那是因为模型

04:16 is trained for significantly longer.
受训时间长得多。

04:17 In this case, 1.4 trillion tokens
在这种情况下,1.4万亿个象征性

04:19 instead of just 300 billion tokens.
而不是3000亿的代币

04:21 So you shouldn't judge the power of a model just
所以你不应该只判断模型的力量

04:23 by the number of parameters that it contains.
以包含的参数数为基准。

04:26 Below, I'm showing some tables of rough hyper-parameters
下面我展示一些粗糙的超参数表

04:30 that typically go into specifying the transformer neural
通常用于指定变压器神经

transformer [træns'fɔ:mə]
Transformer 中文为“变换器”，但不够达雅，日常还是称英文 Transformer 中文为“变换器”，但不够达雅，日常还是称英文
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04:33 network, so the number of heads, the dimension size,
网络,所以头数,尺寸大小

04:36 number of layers, and so on.
数层,等等。

04:37 And on the bottom, I'm showing some training hyper-parameters.
底部,我展示了一些训练超参数

training ['treiniŋ]
训练 确定构成模型的理想参数 的过程。
神经网络基础

04:41 So for example, to train the 65B model,
比如,为了训练65B模型

04:44 Meta used 2,000 GPUs, roughly 21 days of training,
Meta使用了2,000GPUs,大约21天的培训,

training ['treiniŋ]
训练 确定构成模型的理想参数 的过程。
神经网络基础

04:48 and roughly several million dollars.
大约几百万美元

04:52 And so that's the rough orders of magnitude
这就是最粗糙的量级

04:54 that you should have in mind for the pre-training stage.
参加预科培训阶段的人,应当记得这些。

training ['treiniŋ]
训练 确定构成模型的理想参数 的过程。
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04:58 Now, when we're actually pre-training, what happens?
现在,当我们实际上在训练前, 会发生什么?

training ['treiniŋ]
训练 确定构成模型的理想参数 的过程。
神经网络基础

05:01 Roughly speaking, we are going to take our tokens
大致说来,我们是要拿些纪念品

05:03 and we're going to lay them out into data batches.
把它们放入数据组

神经网络 · 概率 05:06

05:06 So we have these arrays that will feed into the transformer.
所以我们有这些阵列 将输入变压器。

transformer [træns'fɔ:mə]
Transformer 中文为"变换器"，但不够达雅，日常还是称英文 Transformer 中文为"变换器"，但不够达雅，日常还是称英文
神经网络基础

05:09 And these arrays are B, the batch size,
这些阵列是B, 批量大小,

05:11 and these are all independent examples stacked up in rows.
这些都是单列的单独例子。

05:15 And B by T, T being the maximum context length.
和B乘以T,T是最大范围长度。

05:18 So in my picture, I only have 10 of the context lengths.
所以在我的照片里,我只有10个背景长度。

05:21 So this could be 2,000, 4,000, et cetera.
也就是说,这可能是2000,4,000,等等。

05:23 So these are extremely long rows.
所以,这些是极为长的排。

05:25 And what we do is we take these documents
我们要做的就是 把这些文件拿走

05:26 and we pack them into rows, and we delimit them
然后,我把他们排成若干行,而加以分

05:29 with these special end of text tokens,
带有这些文本标记的特殊端,

05:31 basically telling the transformer
基本上告诉变压器

transformer [træns'fɔ:mə]
Transformer 中文为"变换器"，但不够达雅，日常还是称英文 Transformer 中文为"变换器"，但不够达雅，日常还是称英文
神经网络基础

05:32 where a new document begins.
中开始新文档。

05:35 And so here I have a few examples of documents,
因此,我这里有一些文件的例子:

05:38 and then I've stretched them out into this input.
然后我就把它们伸展到这个输入中

05:42 Now, we're going to feed all of these numbers
现在,我们要喂饱所有这些数字

05:44 into transformer.
变成变压器。

transformer [træns'fɔ:mə]
Transformer 中文为"变换器"，但不够达雅，日常还是称英文 Transformer 中文为"变换器"，但不够达雅，日常还是称英文
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05:46 And let me just focus on a single particular cell,
让我集中谈谈一个特定的细胞,

05:49 but the same thing will happen at every cell in this diagram.
但同样的事情也会发生在这个图表中的每一个细胞。

05:52 So let's look at the green cell.
让我们看看绿色牢房。

05:54 The green cell is going to take a look
绿色牢房会看一眼

05:56 at all of the tokens before it,
在他们之前,有许多迹象,

05:58 so all of the tokens in yellow.
所以所有的标牌都是黄色的

06:00 And we're going to feed that entire context
并且我们要去喂饱整个环境

06:03 into the transformer neural network.
进入变压器神经网络

transformer [træns'fɔ:mə]
Transformer 中文为"变换器"，但不够达雅，日常还是称英文 Transformer 中文为"变换器"，但不够达雅，日常还是称英文
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neural network
神经网络 一种模型，灵感来源于脑部结构，由多个层构成（至少有一个是隐藏层），每 个层都包含简单相连的单元或神经元 （具有非线性关系）。
神经网络基础

06:05 And the transformer is going to try
变压器将尝试

transformer [træns'fɔ:mə]
Transformer 中文为"变换器"，但不够达雅，日常还是称英文 Transformer 中文为"变换器"，但不够达雅，日常还是称英文
神经网络基础

06:07 to predict the next token in a sequence, in this case, in red.
以红色的顺序预测下一个标记。

06:10 Now, the transformer, I don't have too much time
现在变压器,我没时间了

transformer [træns'fɔ:mə]
Transformer 中文为"变换器"，但不够达雅，日常还是称英文 Transformer 中文为"变换器"，但不够达雅，日常还是称英文
神经网络基础

06:12 to unfortunately go into the full details
不幸地要详细讨论所有细节

06:13 of this neural network architecture,
以及这个神经网络结构的

neural network
神经网络 一种模型，灵感来源于脑部结构，由多个层构成（至少有一个是隐藏层），每 个层都包含简单相连的单元或神经元 （具有非线性关系）。
神经网络基础

06:15 is just a large blob of neural net stuff for our purposes,
仅仅是我们目的上 巨大的神经网材料

neural net
神经网络 神经网络
神经网络基础

06:19 and it's got several 10 billion parameters typically
通常它有几亿个参数

06:21 or something like that.
或类似的东西。

06:22 And of course, as you tune these parameters,
当然,当你调和这些参数时

06:23 you're getting slightly different predicted distributions
预测分布略有不同

06:26 for every single one of these cells.
每一个细胞

06:29 And so, for example, if our vocabulary size is 50,257 tokens,
比如,如果我们的词汇大小是50 257个符号

06:34 then we're going to have that many numbers
然后我们就会有那么多数字

06:36 because we need to specify a probability distribution
因为我们需要指定概率分布

probability
概率 概率
数学与微积分

distribution
分布 分布
数学与微积分

06:39 for what comes next.
未来是什么。

06:40 So basically, we have a probability for whatever may follow.
所以基本上,我们有一个可能性 任何可能随其后。

probability
概率 概率
数学与微积分

06:43 Now, in this specific example for this specific cell,
现在,在这个特定单元格的具体示例中,

06:45 513 will come next.
下一个是513号

06:47 And so we can use this as a source of supervision
以此作为监督的源头

06:49 to update our transformer 's weights .
更新变压器的重量

transformer [træns'fɔ:mə]
Transformer 中文为"变换器"，但不够达雅，日常还是称英文 Transformer 中文为"变换器"，但不够达雅，日常还是称英文
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weights
权重 权重
神经网络基础

06:51 And so we're applying this basically
所以我们基本上应用这个

06:53 on every single cell in parallel,
以每个平行的细胞盟誓,

06:54 and we keep swapping batches, and we're
我们继续交换批次,我们

06:56 trying to get the transformer to make the correct predictions
试图让变压器做出正确的预测

transformer [træns'fɔ:mə]
Transformer 中文为"变换器"，但不够达雅，日常还是称英文 Transformer 中文为"变换器"，但不够达雅，日常还是称英文
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06:59 over what token comes next in a sequence.
象征会先后相继降临

07:01 So let me show you more concretely what this looks like when
所以,让我更具体地告诉你,当

损失函数 · 权重 07:04

07:04 you train one of these models.
你训练了其中的模型之一

07:05 This is actually coming from New York Times.
这其实是纽约时报的

07:08 And they trained a small GPT on Shakespeare.
他们在莎士比亚上训练了一支小GPT

gpt
生成式预训练Transformer模型 生成式预训练Transformer模型
神经网络基础

07:11 And so here's a small snippet of Shakespeare,
莎士比亚的一小块片段

07:12 and they trained a GPT on it.
他们训练了一支GPT

gpt
生成式预训练Transformer模型 生成式预训练Transformer模型
神经网络基础

07:14 Now, in the beginning at initialization,
现在,在初始化的开始阶段

07:17 the GPT starts with completely random weights.
GPT 开始完全随机的重量。

gpt
生成式预训练Transformer模型 生成式预训练Transformer模型
神经网络基础

weights
权重 权重
神经网络基础

07:19 So you're just getting completely random outputs as well.
所以你也只是得到完全随机的产出

07:22 But over time, as you train the GPT longer and longer,
但随着时间的推移,当你训练GPT 越来越长的时间

gpt
生成式预训练Transformer模型 生成式预训练Transformer模型
神经网络基础

07:26 you are getting more and more coherent and consistent samples
越来越多的样本 更加一致一致

07:30 from the model.
模型中的。

07:31 And the way you sample from it, of course,
当然,你从中抽取样本的方式

sample
样本 样本
神经网络基础

07:33 is you predict what comes next.
你是否预知接下来会发生什么?

07:35 You sample from that distribution,
你从分布中抽样

sample
样本 样本
神经网络基础

distribution
分布 分布
数学与微积分

07:36 and you keep feeding that back into the process,
你将那份食物反馈到过程中

07:39 and you can basically sample large sequences.
并且可以基本上抽样 大型序列。

sample
样本 样本
神经网络基础

07:42 And so by the end, you see that the transformer has
所以最后,你可以看到变压器

transformer [træns'fɔ:mə]
Transformer 中文为"变换器"，但不够达雅，日常还是称英文 Transformer 中文为"变换器"，但不够达雅，日常还是称英文
神经网络基础

07:43 learned about words, and where to put spaces,
学会文字, 以及放在哪里空间,

07:46 and where to put commas, and so on.
在哪里放逗号,等等。

07:48 And so we're making more and more consistent predictions
所以我们越来越一致的预测

07:50 over time.
随着时间的推移。

07:51 These are the kinds of plots that you're looking at when
这些是你们在看的地块类型

07:53 you're doing model pre- training .
你在做示范预训练

training ['treiniŋ]
训练 确定构成模型的理想参数 的过程。
神经网络基础

07:54 Effectively, we're looking at the loss function over time
有效看,我们正研究随着时间推移而来的损失功能

loss function
损失函数 损失函数
训练与优化

07:57 as you train.
训练时,

07:58 And low loss means that our transformer
低损失意味着我们的变压器

transformer [træns'fɔ:mə]
Transformer 中文为"变换器"，但不够达雅，日常还是称英文 Transformer 中文为"变换器"，但不够达雅，日常还是称英文
神经网络基础

08:00 is predicting the correct, is giving a higher probability
正在预测正确值, 给的概率更高

probability
概率 概率
数学与微积分

08:03 to get the correct next integer in a sequence.
以获得一个序列中正确的下一个整数。

08:07 Now, what are we going to do with this model
现在,我们该拿这个模型怎么办?

08:09 once we've trained it after a month?
一个月后我们训练过一次?

08:11 Well, the first thing that we noticed, the field,
我们注意到的第一件事, 现场

08:14 is that these models, basically,
这些模型基本上就是

08:16 in the process of language modeling,
正在进行语言模拟,

08:18 learn very powerful general representations.
学会了非常强大的一般表述。

08:21 And it's possible to very efficiently fine-tune them
并且可以非常有效地微调它们

08:23 for any arbitrary downstream task you might be interested in.
任何任意的下游任务 你可能感兴趣。

08:26 So as an example, if you're interested in sentiment
例如,如果你对情绪感兴趣

08:28 classification, the approach used to be
分类、分类、方法、

08:31 that you collect a bunch of positives and negatives,
以便你们收集许多正面和负面的,

08:33 and then you train some kind of an NLP model for that.
然后训练某种NLP模型

08:37 But the new approach is ignore sentiment classification,
但新办法就是忽略情绪分类

08:40 go off and do large language model pre-training,
开始做大型语言模型的预培训

training ['treiniŋ]
训练 确定构成模型的理想参数 的过程。
神经网络基础

08:43 train a large transformer, and then you
火车上一个大变压器,然后你们

transformer [træns'fɔ:mə]
Transformer 中文为"变换器"，但不够达雅，日常还是称英文 Transformer 中文为"变换器"，但不够达雅，日常还是称英文
神经网络基础

08:46 may only have a few examples,
仅举几个例子,

08:47 and you can very efficiently fine-tune your model
并且可以非常高效地微调你的模型

08:49 for that task.
完成这项任务

08:51 And so this works very well in practice.
因此,这个在实践上非常有效。

08:53 And the reason for this is that, basically, the transformer
原因就是, 基本上,变压器

transformer [træns'fɔ:mə]
Transformer 中文为"变换器"，但不够达雅，日常还是称英文 Transformer 中文为"变换器"，但不够达雅，日常还是称英文
神经网络基础

08:55 is forced to multitask a huge amount of tasks
被迫多任务执行大量任务

08:59 in the language modeling task.
语言模型任务。

09:00 Because just in terms of predicting the next token,
因为仅仅在预测下一个信号方面

09:03 it's forced to understand a lot about the structure
被迫了解很多关于结构的

09:05 of the text and all the different concepts therein.
以及其中所有不同的概念。

09:10 So that was GPT -1.
也就是GPT -1

gpt
生成式预训练Transformer模型 生成式预训练Transformer模型
神经网络基础

09:11 Now, around the time of GPT -2,
现在,在GPT-2时段

gpt
生成式预训练Transformer模型 生成式预训练Transformer模型
神经网络基础

09:12 people noticed that actually, even better than fine-tuning,
人们注意到,实际上, 甚至比微调更好,

09:15 you can actually prompt these models very effectively.
您可以非常有效地 激发这些模型。

09:17 So these are language models, and they
这些就是语言模型,它们

09:19 want to complete documents.
想要完成文档。

09:20 So you can actually trick them into performing tasks
所以,你可以让他们 被骗去执行任务

09:23 just by arranging these fake documents.
仅仅通过安排这些假文件

09:25 So in this example, for example, we have some passage,
比如说,在这个例子中,我们有一些通过,

09:28 and then we sort of do QA, QA, QA.
然后我们做 QA,QA, QA。

09:31 This is called a few-shot prompt.
这叫"几发即时信号"

09:32 And then we do Q. And then as the transformer
然后我们做Q,然后当变压器

transformer [træns'fɔ:mə]
Transformer 中文为"变换器"，但不够达雅，日常还是称英文 Transformer 中文为"变换器"，但不够达雅，日常还是称英文
神经网络基础

09:35 is trying to complete the document,
正在试图完成文档,

09:36 it's actually answering our question.
事实上,这是回答我们的问题。

09:37 And so this is an example of prompt engineering, a base
这是快速工程的一个例子,一个基地

09:40 model, making it believe that it's sort of imitating
模型,让它相信 它是一种模仿

09:42 a document and getting it to perform a task.
文档,并获得它来执行任务。

09:45 And so this picked off, I think, the era of, I would
因此,这个被抛弃的时代,我想

09:48 say, prompting over fine-tuning and seeing that this actually
提醒大家过细调 看到这个

09:51 can work extremely well on a lot of problems,
有很多问题可以非常有效

09:53 even without training any neural networks, fine-tuning,
甚至连任何神经网络 都没有经过训练,微调,

training ['treiniŋ]
训练 确定构成模型的理想参数 的过程。
神经网络基础

09:55 or so on.
或等等。

09:57 Now, since then, we've seen an entire evolutionary tree
从那时起,我们看到了一棵进化的树

10:00 of base models that everyone has trained.
每个人所训练的模型

10:03 Not all of these models are available.
并非所有这些模型都可用。

10:05 For example, the GPT -4 base model was never released.
例如,GPT-4基准模型从未公布。

gpt
生成式预训练Transformer模型 生成式预训练Transformer模型
神经网络基础

10:08 The GPT -4 model that you might be interacting with over API
您可能与 API 上的 GPT-4 模式交互

gpt
生成式预训练Transformer模型 生成式预训练Transformer模型
神经网络基础

10:10 is not a base model.
不是一个基准模型。

10:12 It's an assistant model.
这是一个助理模型。

10:13 And we're going to cover how to get those in a bit.
我们准备把那些都埋在后面

10:16 GPT -3 base model is available via the API
GPT-3基准模型可通过API提供

gpt
生成式预训练Transformer模型 生成式预训练Transformer模型
神经网络基础

10:19 under the name DaVinci.
以达文西为名

10:20 And GPT-2 base model is available even as weights
即便在加权数时,GPT-2基准模型也可用

gpt
生成式预训练Transformer模型 生成式预训练Transformer模型
神经网络基础

weights
权重 权重
神经网络基础

10:23 on our GitHub repo.
以我们的吉他神灵盟誓,

10:24 But currently, the best available base model
但目前,最佳可用基准模型

10:27 probably is the Lama series from Meta,
可能是梅塔的拉马系列

10:30 although it is not commercially licensed.
虽然它没有商业许可证。

10:33 Now, one thing to point out is base models are not
现在,有一点要指出的是 基准模型不是

10:36 assistants.
助理。

训练集 · 数据集 10:36

10:36 They don't want to make answers to your questions.
他们不想回答你的问题

10:41 They just want to complete documents.
他们只想完成文件

10:43 So if you tell them write a poem about a brand cheese,
如果你告诉他们写一首关于品牌奶酪的诗

10:46 it will answer questions with more questions.
它将回答问题,并提出更多问题。

10:49 It's just completing what it thinks is a document.
它只是完成了它所认为的文件

10:51 However, you can prompt them in a specific way
不过,您可以特定方式催化

10:53 for base models that is more likely to work.
用于更可能发挥作用的模型。

10:57 So as an example, here's a poem about bread and cheese.
比如,这里有首关于面包和奶酪的诗。

10:59 And in that case, it will autocomplete correctly.
如果这样的话,它就会完全自动完成

11:02 You can even trick base models into being assistants.
你甚至可以欺骗模型 成为助理。

11:06 And the way you would do this is
你会这样做的方式是

11:07 you would create a specific few shot prompt that
将创建一个特定的小镜头提示

11:10 makes it look like there's some kind of a document
使文件看起来像有文档

11:12 between a human and assistant, and they're exchanging
人与助理,他们交换

11:14 sort of information.
信息排序。

11:16 And then at the bottom, you sort of put your query at the end.
然后在底部,你把查询放在结尾。

11:19 And the base model will sort of condition itself
基模本身也会有某种条件

11:23 into being like a helpful assistant and answer.
成为帮助的助手和回答

11:26 But this is not very reliable and doesn't work super well
但这不可靠 工作效果也不好

11:28 in practice, although it can be done.
在实践中,尽管可以做到。

11:31 So instead, we have a different path
所以,我们有一条不同的路

11:32 to make actual GPT assistants, not just base model,
使全球电话网实际成为全球电话网的助理,而不仅仅是基础模式,

gpt
生成式预训练Transformer模型 生成式预训练Transformer模型
神经网络基础

11:35 document completers.
文档完整器。

11:37 And so that takes us into supervised fine tuning.
这使得我们进入了监管的微调。

fine tuning
微调 微调
神经网络基础

11:39 So in the supervised fine tuning stage,
所以在监督的微调阶段

fine tuning
微调 微调
神经网络基础

11:41 we are going to collect small but high quality data
我们将会收集小但高质量的数据

11:44 sets.
设置。

11:45 And in this case, we're going to ask human contractors
此时,我们请人来做承包商

11:47 to gather data of the form prompt and ideal response.
收集表格数据,迅速和理想地作出反应。

11:52 And we're going to collect lots of these, typically tens
我们收集了很多这些,通常有几十个

11:54 of thousands or something like that.
成千上万或类似的东西。

11:56 And then we're going to still do language modeling
然后我们还会继续做语言模型

11:58 on this data.
数据。

11:59 So nothing changed algorithmically.
所以没有改变任何逻辑

12:01 We're just swapping out a training set.
我们只是换个训练场面

training ['treiniŋ]
训练 确定构成模型的理想参数 的过程。
神经网络基础

12:02 So it used to be internet documents, which
所以,它过去曾经是互联网文件,

12:04 is a high quantity local for basically QA prompt response
基本上对质量评估迅速反应而言,是当地高数量的数量

12:10 kind of data.
数据类型。

12:11 And that is low quantity, high quality.
数量少,质量高

12:14 So we will still do language modeling.
因此,我们仍将进行语言模型的模拟。

12:15 And then after training, we get an SFD model.
训练后,我们得到了一个SFD模型

training ['treiniŋ]
训练 确定构成模型的理想参数 的过程。
神经网络基础

12:18 And you can actually deploy these models.
你可以实际部署这些模型。

12:20 And they are actual assistants.
他们是真正的助理

12:21 And they work to some extent.
并且它们在某种程度上是有效的。

12:23 Let me show you what an example demonstration might
让我来示范一下

12:25 look like.
看起来像。

12:25 So here's something that a human contractor might come up with.
所以,这是人类承包商可能想出的东西

12:28 Here's some random prompt.
这儿有些随机的提示

12:29 Can you write a short introduction
您可写一个简短的介绍

12:31 about the relevance of the term monopsony
单一性一词的相关性

12:33 or something like that?
或类似的东西?

12:34 And then the contractor also writes out an ideal response.
然后承包商也写出一个理想的反应

12:37 And when they write out these responses,
当他们写下这些答复时,

12:38 they are following extensive labeling documentations.
他们正在跟踪大量标签文件。

12:41 And they are being asked to be helpful, truthful, and harmless.
他们将被召去行善,说情,说情而无害。

12:45 And those labeling instructions here,
还有标签上的指示

12:48 you probably can't read it.
你可能读不出来

12:49 Neither can I.
我也不能

12:50 But they're long.
但他们长。

12:52 And this is just people following instructions
而这只是人遵循指示

12:53 and trying to complete these prompts.
并努力完成这些提示。

12:55 So that's what the data set looks like.
所以数据组就是这个样子的

data set
数据集 一组样本 的集合。 Dataset API (tf.data) 一种高级别的 TensorFlow API ，用于读取数据并将其转换为机器学习算法所需的格式。 tf.data.Dataset 对象表示一系列元素，其中每个元素都包含一个或多个张量 。 tf.data.Iterator 对象可获取 Dataset 中的元素。 如需详细了解 D
神经网络基础

人类反馈的强化学习 · 强化学习 12:57

12:57 And you can train these models.
你可以训练这些模型

12:58 And this works to some extent.
在某种程度上,这在一定程度上是有效的。

13:00 Now, you can actually continue the pipeline from here on
现在,你可以继续从这里开始

13:03 and go into RLHF, Reinforcement Learning
并进入RLHF, 强化学习

rlhf
人类反馈的强化学习 人类反馈的强化学习
神经网络基础

reinforcement learning
强化学习 强化学习
神经网络基础

13:06 from Human Feedback.
人类反馈。

13:07 That consists of both reward modeling
包括两个奖赏模式

13:09 and reinforcement learning.
并加强学习。

reinforcement learning
强化学习 强化学习
神经网络基础

13:10 So let me cover that.
让我来覆盖一下

13:11 And then I'll come back to why you
然后我会回来找你

13:12 may want to go through the extra steps and how that compares
可能想要通过额外的步骤和如何比较

13:14 to just SFD models.
仅限SFD模型。

13:16 So in the reward modeling step, what we're going to do
所以在奖励的模范阶段, 我们要做的是

13:19 is we're now going to shift our data collection
也就是说我们现在要转移数据收集

13:20 to be of the form of comparisons.
以比较的形式。

13:23 So here's an example of what our data set will look like.
下面是一个例子,我们的数据组会是什么样子。

data set
数据集 一组样本 的集合。 Dataset API (tf.data) 一种高级别的 TensorFlow API，用于读取数据并将其转换为机器学习算法所需的格式。 tf.data.Dataset 对象表示一系列元素，其中每个元素都包含一个或多个张量 。 tf.data.Iterator 对象可获取 Dataset 中的元素。 如需详细了解 D
神经网络基础

13:25 I have the same prompt, identical prompt on the top,
我上面有同样的快速,相同的速度

13:28 which is asking the assistant to write a program or a function
要求助理撰写程序或函数

13:32 that checks if a given string is a palindrome.
检查给定字符串是否为暗淡区。

13:35 And then what we do is we take the SFD model, which we've
然后我们要做的就是 采用SFD模式,我们有

13:38 already trained.
已经培训。

13:39 And we create multiple completions.
我们创造出多重完成。

13:41 So in this case, we have three completions
所以,在这种情况下,我们有三个完成

13:42 that the model has created.
模型已经创造了。

13:44 And then we ask people to rank these completions.
然后我们要求人们排出这些结业的排名

13:47 So if you stare at this for a while,
如果你盯着这个看一阵子

13:48 and by the way, these are very difficult things
顺便说一句,这些是困难的事情

13:50 to do to compare some of these predictions,
来比较其中一些预测,

13:52 and this can take people even hours
这甚至可以花费人的时间

13:54 for single prompt completion pairs.
单速成品的双对。

13:58 But let's say we decided that one of these
但是我们决定 其中一个

14:00 is much better than the others and so on.
比其他人好得多等等

14:02 So we rank them.
所以我们排出他们的位置

14:03 Then we can follow that with something
然后我们可以跟着它 与一些东西

14:04 that looks very much like a binary classification
看起来像二进制分类

14:07 on all the possible pairs between these completions.
所有的可能的对 之间的这些完成。

14:10 So what we do now is we lay out our prompt in rows,
所以我们现在要做的是 排成一行地展示我们的迅速性

14:13 and the prompts is identical across all three rows here.
三个行的提示都是一样的

14:16 So it's all the same prompt, but the completion is very.
所以,这都是相同的 及时,但完成是非常非常。

14:19 And so the yellow tokens are coming from the SFD model.
所以黄色的象征品 是从SFD模型中传来的

14:22 Then what we do is we append another special reward
那我们就只做附加另一个特别奖励

14:25 readout token at the end, and we basically only
结尾处的读出标记, 我们基本上只

14:28 supervise the transformer at this single green token.
监督变压器的这个绿色标志。

transformer [træns'fɔ:mə]
Transformer 中文为"变换器"，但不够达雅，日常还是称英文 Transformer 中文为"变换器"，但不够达雅，日常还是称英文
神经网络基础

14:31 And the transformer will predict some reward
变压器将预言一些报酬

transformer [træns'fɔ:mə]
Transformer 中文为"变换器"，但不够达雅，日常还是称英文 Transformer 中文为"变换器"，但不够达雅，日常还是称英文
神经网络基础

14:34 for how good that completion is for that prompt.
完成这个任务对于迅速完成这个任务是多么好。

14:38 And so basically it makes a guess about the quality
所以基本上它能猜出质量

14:41 of each completion.
完成的每份。

14:42 And then once it makes a guess for every one of them,
一旦它让每个猜想出来

14:44 we also have the ground truth, which
我们也有事实真相,

14:46 is telling us the ranking of them.
告诉我们他们排名的顺序

14:48 And so we can actually enforce that some of these numbers
所以我们可以执行其中的一些数字

14:50 should be much higher than others and so on.
其数量应该比其他人高得多等等。

14:52 We formulate this into a loss function,
我们把它变成一种损失功能,

loss function
损失函数 损失函数
训练与优化

14:54 and we train our model to make reward predictions that
我们训练我们的模型 作出奖励预测

14:56 are consistent with the ground truth coming
符合事实真相的真相

14:58 from the comparisons from all these contractors.
与所有这些承包者的比较。

15:01 So that's how we train a reward model.
所以我们训练奖赏模式

15:02 And that allows us to score how good a completion is
从而让我们能分得好成绩

15:05 for a prompt.
快速的

15:07 Once we have a reward model, we can't deploy this
一旦我们有了奖赏模式, 我们就不能部署这个

15:10 because this is not very useful as an assistant by itself,
因为这本身作为一名助理并没有多大用处,

15:13 but it's very useful for the reinforcement learning
但它对强化学习非常有用

reinforcement learning
强化学习 强化学习
神经网络基础

15:15 stage that follows now.
之后的阶段。

15:17 Because we have a reward model, we
因为我们有奖赏模式

15:18 can score the quality of any arbitrary completion
任何任意完成的成绩都可分得质量

15:21 for any given prompt.
以便他们迅速获得。

15:23 So what we do during reinforcement learning
所以,我们在强化学习期间做什么

reinforcement learning
强化学习 强化学习
神经网络基础

15:24 is we basically get, again, a large collection of prompts,
我们基本上还是能再次得到大量的提示

15:27 and now we do reinforcement learning with respect
现在我们在加强学习的同时尊重

reinforcement learning
强化学习 强化学习
神经网络基础

15:29 to the reward model.
至于奖赏模式。

15:30 So here's what that looks like.
情况是这样的

15:32 We take a single prompt, we lay it out in rows,
我们只用一瞬间,我们排成若干行,

15:36 and now we use the SFD model.
现在我们使用SFD模型

15:37 We use basically the model we'd like to train,
我们基本上使用我们想训练的模型

15:39 which is initialized at SFD model,
以SFD模型初始化,

15:41 to create some completions in yellow.
创建一些黄色的补全。

15:43 And then we append the reward token again,
然后,我们又附上奖赏的记念,

15:46 and we read off the reward according to the reward model,
我这样以报酬的态度,宣读报酬,

15:49 which is now kept fixed.
现已固定。

15:50 It doesn't change anymore.
已经没有变化了

15:52 And now the reward model tells us
现在奖励模式告诉我们

15:54 the quality of every single completion for these prompts.
每一个完成这些提示的质量。

15:57 And so what we can do is we can now just basically
所以我们所能做的就是 我们现在可以基本上

15:59 apply the same language modeling loss function,
适用相同的语言模拟损失功能,

loss function
损失函数 损失函数
训练与优化

16:01 but we're currently training on the yellow tokens,
但我们目前正在训练黄色的标志

training ['treinɪŋ]
训练 确定构成模型的理想参数 的过程。
神经网络基础

16:04 and we are weighing the language modeling objective
并且我们正权衡语言模型的目标

16:08 by the rewards indicated by the reward model.
以表率所显示的,是报酬。

16:11 So as an example, in the first row,
例如,第一行

16:13 the reward model said that this is a fairly high scoring
奖赏模式说,这是一个相当高的评分

16:16 completion.
完成。

16:17 And so all of the tokens that we happened to sample
所以所有我们碰巧抽样的象征

sample
样本 样本
神经网络基础

16:20 on the first row are going to get reinforced,
第一排的加固

16:22 and they're going to get higher probabilities for the future.
将来他们就会有更高的概率

16:25 Conversely, on the second row, the reward model really
反之,第二排的奖赏模式真的

16:27 did not like this completion, negative 1.2.
并不喜欢完成这一完成,负1.2。

16:30 And so therefore, every single token
因此,每个象征

16:32 that we sampled in that second row
并在第二行中取样

16:33 is going to get a slightly higher probability for the future.
未来的可能性会略高一点

probability
概率 概率
数学与微积分

16:36 And we do this over and over on many prompts, on many batches,
我们反复地做这些 在许多提示, 许多批次的,

16:39 and basically, we get a policy which creates yellow tokens
基本上,我们得到一个政策 创造黄色的象征物

16:42 here, and basically all of the completions here
这里,基本上这里所有的完成

16:46 will score high according to the reward model that we
得高分的奖赏模式 我们

16:48 trained in the previous stage.
上阶段受过培训。

16:51 So that's how we train.
所以我们就是这么训练的

16:53 That's what the RLHF pipeline is.
RLHF管道就是这样的

rlhf
人类反馈的强化学习 人类反馈的强化学习
神经网络基础

16:56 And then at the end, you get a model that you could deploy.
最终,你得到一个可以部署的模型

16:59 And as an example, chatGPT is an RLHF model.
例如,聊天GPT是一种RLHF模式。

> **rlhf**
> 人类反馈的强化学习 人类反馈的强化学习
> 神经网络基础

17:02 But some other models that you might come across,
但有些其他的模型 你可能会遇到,

17:04 like for example, the KUNA 13b and so on,
例如,KUNA 13b等等

17:06 these are SFD models.
这些是SFD模型。

17:08 So we have base models, SFD models, and RLHF models.
因此,我们有基准模型、SFD模型和RLHF模型。

> **rlhf**
> 人类反馈的强化学习 人类反馈的强化学习
> 神经网络基础

17:11 And that's kind of like the state of things there.
有点像那里的情况

17:14 Now why would you want to do RLHF?
现在,你为什么要做RLHF?

> **rlhf**
> 人类反馈的强化学习 人类反馈的强化学习
> 神经网络基础

17:17 So one answer that is kind of not that exciting
所以一个答案 并不那么刺激

17:19 is that it just works better.
因为它只是工作更好。

17:21 So this comes from the InstructGPT paper.
所以这是来自 指令GPT纸。

17:23 According to these experiments a while ago now,
根据这些实验 不久前,现在,

17:25 these PPO models are RLHF.
这些PPO模式是RLHF。

> **rlhf**
> 人类反馈的强化学习 人类反馈的强化学习
> 神经网络基础

17:28 And we see that they are basically just preferred
我们看到他们基本上只是被偏爱

17:30 in a lot of comparisons when we give them to humans.
和人类的对比

17:33 So humans just prefer basically tokens that come from RLHF
所以人类喜欢的是来自RLHF的 基本象征物

> **rlhf**
> 人类反馈的强化学习 人类反馈的强化学习
> 神经网络基础

17:37 models compared to SFD models, compared to base model that
与SFD模型相比,比较了SFD模式的模型,对比了基准模型,而基准模型

17:40 is prompted to be an assistant.
被激励做助手。

训练数据 · 熵 17:42

17:42 And so it just works better.
所以效果更好了

17:43 But you might ask why?
但你可能会问为什么?

17:46 Why does it work better?
为什么它的工作效果更好?

17:47 And I don't think that there's a single amazing answer
我不认为有一个 令人惊叹的答案

17:49 that the community has really agreed on.
社会真正同意了这一点。

17:51 But I will just offer one reason potentially.
但我会提出一个可能的理由

17:55 And it has to do with the asymmetry between how easy
和它必须做 与不对称的

17:58 computationally it is to compare versus generate.
计算它是比较和生成。

18:02 So let's take an example of generating a haiku.
让我们举个例子来做个创造大名词的例子

18:05 Suppose I ask a model to write a haiku about paper clips.
假设我要求一个模型来写一篇关于纸片的口述

18:08 If you're a contractor trying to give training data,
如果你是想提供训练数据的承包商

training ['treinin]
训练 确定构成模型的理想参数 的过程。
神经网络基础

18:10 then imagine being a contractor collecting basically
然后想象成为承包商 收集基本

18:12 data for the SFD stage.
供SFD阶段使用的数据。

18:14 How are you supposed to create a nice haiku
怎样才能造出一个好听的白道

18:16 for a paper clip?
纸片?

18:16 You might just not be very good at that.
你也许不擅长这个

18:18 But if I give you a few examples of haikus,
但如果我给你几个关于哈伊库斯的例子

18:20 you might be able to appreciate some of these haikus
可能你能够欣赏到一些这些哈伊科斯

18:22 a lot more than others.
比别人要多得多

18:24 And so judging which one of these is good
判断其中哪一种是好的

18:25 is a much easier task.
这是一项容易得多的任务。

18:27 And so basically, this asymmetry makes
所以基本上,这种不对称使得

18:29 it so that comparisons are a better way
以便比较是更好的方法

18:32 to potentially leverage yourself as a human and your judgment
有可能利用自己作为人 和你的判断力

18:35 to create a slightly better model.
创建略为更好的模型。

18:38 Now, our large of models are not strictly an improvement
现在,我们庞大的模型 严格来说并不是改进

18:41 on the base models in some cases.
在某些情况下,以基准模型为基础。

18:43 So in particular, we'd notice, for example,
特别是,我们注意到,例如

18:45 that they lose some entropy.
并且失去一些银河系

entropy
熵 熵
数学与微积分

18:46 So that means that they give more peaky results.
这意味着它们带来更多令人沮丧的结果。

18:49 They can output lower variations.
它们可以产生较低的变异。

18:53 They can output samples with lower variation than base model.
它们可以输出比基模变差小的样品。

18:56 So base model has lots of entropy,
所以模型基础有很多的 entropy,

entropy
熵 熵
数学与微积分

18:57 and will give lots of diverse outputs.
并且有许多不同的产出。

19:00 So for example, one kind of place where I still
比如说,我仍然在某处

19:03 prefer to use a base model is in a setup
使用基模式是首选的设置

19:06 where you basically have n things,
在那里你基本上有东西,

19:10 and you want to generate more things like it.
并且想创造更多类似的东西。

人类反馈的强化学习 · 权重 19:13

19:13 And so here is an example that I just cooked up.
举个例子,我刚刚做了一个。

19:16 I want to generate cool Pokemon names.
我想产生酷的波克蒙名字

19:18 I gave it seven Pokemon names, and I asked the base model
我给它七个波克蒙名字,我问了基模

19:21 to complete document.
以完成文档。

19:22 And it gave me a lot more Pokemon names.
给了我更多波科蒙名字

19:24 These are fictitious.
这些是虚构的。

19:26 I tried to look them up.
我试着查了一下

19:27 I don't believe they're actual Pokemons.
我不相信他们是真正的波科蒙人

19:29 And this is the kind of task that I think base model would
这正是我认为基准模型

19:31 be good at, because it still has lots of entropy,
因为它仍然有很多的 环球,

entropy
熵 熵
数学与微积分

19:33 and will give you lots of diverse, cool, kind of more
并赏赐你们许多多样的、酷的、更优美的

19:36 things that look like whatever you give it before.
事物看起来像你以前给它的东西

19:40 So this is what, having said all that,
所以,在说了这么多之后,

19:43 these are kind of like the assistant models that are
这些是某种类似助理模型的

19:45 probably available to you at this point.
此时此刻你可能可以使用

19:47 There's a team at Berkeley that ranked a lot of the
伯克利有一支队伍 排在

19:50 available assistant models, and gave them basically
现有助理模型,并基本提供给

19:52 the low ratings.
低收视率。

19:53 So currently some of the best models, of course,
所以目前一些最好的模型 当然

19:55 are GPT -4, by far, I would say, followed by Claude,
远至现在,我可以说是GPT-4, 紧随其后是克劳德

gpt
生成式预训练Transformer模型 生成式预训练Transformer模型
神经网络基础

19:58 GPT -3.5, and then a number of models.
GPT-3.5,然后是若干模型。

gpt
生成式预训练Transformer模型 生成式预训练Transformer模型
神经网络基础

20:00 Some of these might be available as weights,
其中一些可能作为权重,

weights
权重 权重
神经网络基础

20:02 like the Kuna, Koala, et cetera.
库纳人,科阿拉人等等。

20:04 And the first three rows here, they are all
头三排的排在这里,他们都是

20:08 RLHF models, and all of the other models, to my knowledge,
据我所知,RLHF模型 和所有其他模型

rlhf
人类反馈的强化学习 人类反馈的强化学习
神经网络基础

20:11 are SFD models, I believe.
我相信是SFD模型

20:16 OK, so that's how we train these models on a high level.
OK,所以这就是我们如何训练这些模型 在高水平。

20:19 Now I'm going to switch gears, and let's look at how we
现在我要换换换换档 让我们看看我们

20:21 can best apply a GPT-assisted model to your problems.
最好能将GPT辅助模式适用于您的问题。

> **gpt**
> 生成式预训练Transformer模型 生成式预训练Transformer模型
> 神经网络基础

20:26 Now I would like to work in setting up a concrete example.
现在,我想努力树立一个具体的例子。

20:30 So let's work with a concrete example here.
因此,让我们在这里以具体的例子来开展工作。

20:33 Let's say that you are working on an article or a blog post,
假设你正在写文章或博客文章

20:35 and you're going to write this sentence at the end.
并且你将在结尾写下这句话

20:38 California's population is 53 times that of Alaska.
加利福尼亚的人口是阿拉斯加的53倍

20:41 So for some reason you want to compare the populations
所以出于某种原因,你想要比较人口

20:43 of these two states.
这两个国家。

20:45 Think about the rich internal monologue and tool use,
想想丰富的内部独白和工具使用

20:48 and how much work actually goes computationally
以及实际工作有多少在计算

20:50 in your brain to generate this one final sentence.
脑中产生这句最后一句

20:53 So here's maybe what that could look like in your brain.
所以也许这就是你的大脑里 可能长得像什么

20:55 OK, for this next step, let me compare these two populations.
下一步,让我比较一下这两个人口。

21:01 OK, first I'm going to obviously
好,我首先要明显

21:02 need to get both of these populations.
需要同时得到这两个人口。

21:05 Now I know that I probably don't know these populations
现在我知道我可能不认识这些人

21:07 off the top of my head.
在我的头上。

21:09 So I'm kind of like aware of what I know or don't know
所以我有点了解我所知道的

21:11 of my self-knowledge, right?
自我认知,对不对?

21:13 So I do some tool use, and I go to Wikipedia,
所以我用一些工具,然后去维基百科

21:16 and I look up California's population
我查加州的人口

21:18 and Alaska's population.
阿拉斯加的人口也都这样

21:20 Now I know that I should divide the two.
现在我知道我应该把两者分开了

21:22 But again, I know that dividing 39.2 by 0.74
但同样 我知道39.2除以0.74

21:25 is very unlikely to succeed.
不太可能成功。

21:27 That's not the kind of thing that I can do in my head.
我脑海里可不能做这种事

21:29 And so therefore I'm going to rely on a calculator.
因此我将依靠计算器

21:32 So I'm going to use a calculator, punch it in,
所以我要用计算器 插进去

21:34 and see that the output is roughly 53.
并且看到产出大约是53。

21:37 And then maybe I do some reflection and sanity checks
然后也许我做一些反思 和理智检查

21:40 in my brain.
在我的大脑里。

21:40 So does 53 make sense?
53号合情合理吗?

21:42 Well, that's quite a large fraction,
相当大一部分

21:44 but then California has the most popular state,
但加利福尼亚州却最流行

21:46 so maybe that looks OK.
所以也许看起来还行

21:48 So then I have all the information I might need,
所以我就掌握了所有我可能需要的信息

21:50 and now I get to the sort of creative portion of writing.
现在我开始写作的创意部分了

21:52 So I might start to write something like, California has 53x times
所以我可能开始写一些东西 比如加州有53次

21:56 greater, and then I think to myself, that's actually
更强烈,然后我想到我自己, 这实际上是

21:59 like really awkward phrasing.
像是很尴尬的语句

22:00 So let me actually delete that, and let me try again.
因此,让我删除这一段,让我再试一次。

22:03 And so as I'm writing, I have this separate process almost
写作时,我几乎有这个不同的过程

22:06 inspecting what I'm writing and judging
检查我写作和判断的内容

22:09 whether it looks good or not.
不论看起来好坏

22:11 And then maybe I delete, and maybe I reframe it,
然后也许我删除, 也许我重新定义它,

22:13 and then maybe I'm happy with what comes out.
之后也许我对结果很满意

22:15 So basically long story short, a ton
简说长话,一吨

22:17 happens under the hood in terms of your internal monologue
发生在头罩下 以你内部独白为名

22:20 when you create sentences like this.
当你这样发音的时候

22:22 But what does a sentence like this
但这种句子是干什么的

词块 · 样本 22:23

22:23 look like when we are training a GPT on it?
看起来像我们训练GPT吗?

training ['treinɪŋ]
训练 确定构成模型的理想参数 的过程。
神经网络基础

gpt
生成式预训练Transformer模型 生成式预训练Transformer模型
神经网络基础

22:27 From GPT 's perspective, this is just a sequence of tokens.
GPT认为,这只是一系列的象征。

gpt
生成式预训练Transformer模型 生成式预训练Transformer模型
神经网络基础

22:30 So a GPT, when it's reading or generating these tokens,
所以GPT,当它读取或产生这些标志时

gpt
生成式预训练Transformer模型 生成式预训练Transformer模型
神经网络基础

22:34 it just goes chunk, chunk, chunk, chunk, chunk.
块块块块,块块块、块块块

chunk [tʃʌŋk]
词块 词块
神经网络基础

22:36 And each chunk is roughly the same amount
每个块块的数量大致相同

chunk [tʃʌŋk]
词块 词块
神经网络基础

22:38 of computational work for each token.
每件物品的计算工作。

22:40 And these transformers are not very shallow networks.
这些变压器不是浅的网络

22:43 They have about 80 layers of reasoning,
他们有80层推理,

22:45 but 80 is still not too much.
但80个还不算太多。

22:47 And so this transformer is going to do its best to imitate.
所以这个变压器将尽力模仿

transformer [træns'fɔ:mə]
Transformer 中文为"变换器"，但不够达雅，日常还是称英文 Transformer 中文为"变换器"，但不够达雅，日常还是称英文
神经网络基础

22:51 But of course, the process here looks very, very different
当然,这里的进程看起来 与过去完全不同

22:54 from the process that you took.
是从你所选择的过程里

22:56 So in particular, in our final artifacts,
特别是,在我们最后的文物中

22:59 in the data sets that we create
中创建的数据集

23:00 and then eventually feed to LLMs,
然后,然后,你们把雨水供养给大地上的所有大麦,

23:02 all of that internal dialogue is completely stripped.
内部对话全部被完全剥夺。

23:04 And unlike you, the GPT will look at every single token
不像你,GPT会看每件东西

gpt
生成式预训练Transformer模型 生成式预训练Transformer模型
神经网络基础

23:09 and spend the same amount of compute on every one of them.
并花费同样的计算量 在每一个。

23:12 And so you can't expect it to actually like,
所以你不能指望它会喜欢

23:14 well, you can't expect it to do too much work per token.
你不能指望它做太多的活

23:19 So, and also in particular,
所以,还有 特别是

23:21 basically these transformers are just token simulators.
这些变压器基本上只是象征性的模拟器

23:23 So they don't know what they don't know,
所以他们不知道什么 他们不知道,

23:25 like they just imitate the next token.
好像他们仿照下一批象征一样。

23:28 They don't know what they're good at or not good at.
他们不知道自己擅长什么 或不擅长什么

23:29 They just tried their best to imitate the next token.
他们只是尽力模仿下一招

23:32 They don't reflect in the loop.
他们不会在循环中反射

23:34 They don't sanity check anything.
他们什么也检查不出来

23:35 They don't correct their mistakes along the way by default.
他们不会在不履行合同的情况下纠正错误

23:38 They just sample token sequences.
他们只是抽样象征性序列

sample
样本 样本
神经网络基础

23:41 They don't have separate inner monologue streams
他们没有不同的 独白的内分化

23:43 in their head, right?
他们的头,对不对?

算法 · 大语言模型 23:43

23:43 They are evaluating what's happening.
他们在评估发生的事情

23:45 Now, they do have some sort of cognitive advantages,
现在,他们确实有某种认知优势

23:48 I would say, and that is that they do actually have
我想说,这就是说,他们确实确实

23:51 a very large fact-based knowledge
大量基于事实的知识

23:52 across a vast number of areas
横跨众多领域

23:54 because they have, say, several 10 billion parameters.
因为它们有,比如说,有几100亿参数。

23:57 So it's a lot of storage for a lot of facts.
所以它有很多储存 大量的事实。

24:00 But, and they also, I think,
但是,他们也,我想

24:01 have a relatively large and perfect working memory.
具有相对大和完美的工作记忆。

24:04 So whatever fits into the context window
所以无论适合上下文窗口

24:08 is immediately available to the transformer
变压器可立即使用

> **transformer** [træns'fɔ:mə]
> Transformer 中文为"变换器"，但不够达雅，日常还是称英文 Transformer 中文为"变换器"，但不够达雅，日常还是称英文
> 神经网络基础

24:10 through its internal self-attention mechanism.
通过其内部自留机制。

24:12 And so it's kind of like perfect memory,
所以它有点像完美的记忆

24:15 but it's got a finite size.
但它的尺寸是有限的

24:16 But the transformer has a very direct access to it,
但变压器可以直接进入它

> **transformer** [træns'fɔ:mə]
> Transformer 中文为"变换器"，但不够达雅，日常还是称英文 Transformer 中文为"变换器"，但不够达雅，日常还是称英文
> 神经网络基础

24:18 and so it can, like, losslessly remember anything
以便它能一样, 忘却一切

24:21 that is inside its context window.
在其上下文窗口中。

24:24 So that's kind of how I would compare those two.
所以我会比较这两个人

24:25 And the reason I bring all of this up
以及我之所以提出这一切

24:27 is because I think to a large extent,
因为我在很大程度上认为

24:29 prompting is just making up for this sort of cognitive
催促只是弥补了这种认知

24:33 difference between these two kind of architectures,
这两种建筑的区别

24:37 like our brains here and LLM brains.
就像我们在这里的大脑和LLM大脑。

llm
大语言模型 大语言模型
神经网络基础

24:39 You can look at it that way almost.
你可以这样看它,几乎。

24:42 So here's one thing that people found, for example,
举例来说,这里有一件事 人们发现了

24:44 works pretty well in practice.
实际工作相当顺利。

24:45 Especially if your tasks require reasoning,
尤其是如果你的任务需要理性

24:48 you can't expect the transformer to make,
你不能指望变压器能造出来

transformer [træns'fɔ:mə]
Transformer 中文为“变换器”，但不够达雅，日常还是称英文 Transformer 中文为“变换器”，但不够达雅，日常还是称英文
神经网络基础

24:50 to do too much reasoning per token.
做太多的推理 每一个象征性。

24:52 And so you have to really spread out the reasoning
所以你必须把推理的

24:54 across more and more tokens.
越来越多的象征

24:56 So for example, you can't give a transformer
比如说,你不能给变压器

transformer [træns'fɔ:mə]
Transformer 中文为"变换器"，但不够达雅，日常还是称英文 Transformer 中文为"变换器"，但不够达雅，日常还是称英文
神经网络基础

24:57 a very complicated question
一个非常复杂的问题

24:59 and expect it to get the answer in a single token.
并期望它得到答案 一个单一的象征性。

25:01 There's just not enough time for it.
只是时间不够

25:02 These transformers need tokens to think,
这些变压器需要用象征来思考

25:04 quote unquote, I like to say sometimes.
引自引文,我有时会说

25:06 And so this is some of the things that work well.
这就是一些工作效果很好的东西。

25:09 You may, for example, have a few shot prompt
例如,您可以使用几发快速

25:10 that shows the transformer that it should show its work
显示变压器应该显示其工作

transformer [træns'fɔ:mə]
Transformer 中文为"变换器"，但不够达雅，日常还是称英文 Transformer 中文为"变换器"，但不够达雅，日常还是称英文
神经网络基础

25:13 when it's answering the question.
回答问题的时候

25:16 And if you give a few examples,
如果你举几个例子

25:17 the transformer will imitate that template
变压器将仿制该模板

transformer [træns'fɔ:mə]
Transformer 中文为"变换器"，但不够达雅，日常还是称英文 Transformer 中文为"变换器"，但不够达雅，日常还是称英文
神经网络基础

25:19 and it will just end up working out better
最终会更好

25:22 in terms of its evaluation.
就其评价而言。

25:24 Additionally, you can elicit this kind of behavior
此外,你可以诱发这种行为

25:26 from the transformer by saying, let's think step by step
说,让我们一步一步地思考

transformer [træns'fɔ:mə]
Transformer 中文为"变换器"，但不够达雅，日常还是称英文 Transformer 中文为"变换器"，但不够达雅，日常还是称英文
神经网络基础

25:29 because this conditioned the transformer
因为这给变压器设置了条件

transformer [træns'fɔ:mə]
Transformer 中文为"变换器"，但不够达雅，日常还是称英文 Transformer 中文为"变换器"，但不够达雅，日常还是称英文
神经网络基础

25:31 into sort of like showing its work.
类似展示它的工作。

25:33 And because it kind of snaps into a mode
因为它会突然进入一种模式

25:35 of showing its work,
显示其工作,

25:36 it's going to do less computational work per token.
它将减少每份象征性的计算工作。

25:40 And so it's more likely to succeed as a result
因此,它更有可能成功 因而

25:42 because it's making slower reasoning over time.
因为它在慢慢推理

25:46 Here's another example.
还有一个例子

25:47 This one is called self-consistency.
这叫自相矛盾

25:50 We saw that we had the ability to start writing
我们看到我们有能力开始写作

25:52 and then if it didn't work out, I can try again
如果结果不成功 我可以再试一次

25:55 and I can try multiple times
我可以尝试多次

25:57 and maybe select the one that worked best.
也许选择一个最成功的人

26:00 So in these kinds of approaches,
所以在这种方式中,

26:02 you may sample not just once,
不仅一次取样,

sample
样本 样本
神经网络基础

26:03 but you may sample multiple times
,但您可以多次取样

sample
样本 样本
神经网络基础

26:05 and then have some process
然后有一个进程

26:06 for finding the ones that are good
寻找好的东西

26:08 and then keeping just those samples
然后只保存这些样本

26:10 or doing a majority vote or something like that.
或者做多数投票之类的事

26:12 So basically these transformers in the process
所以基本上这些变压器 正在运行中

26:14 as they predict the next token, just like you,
他们预测下一个征兆 就像你一样

26:17 they can get unlucky
运气不好的人

26:18 and they could sample a not a very good token
他们可以采一个非善兆,

sample
样本 样本
神经网络基础

26:20 and they can go down sort of like a blind alley
并且他们可以像盲巷一样倒下

26:22 in terms of reasoning.
推理方面。

26:24 And so unlike you, they cannot recover from that.
和你不同,他们无法从中恢复

26:27 They are stuck with every single token they sample
采样的每件物品都粘在他们身上

sample
样本 样本
神经网络基础

26:29 and so they will continue the sequence
以继续顺序

26:31 even if they even know
即使他们知道

26:32 that this sequence is not going to work out.
这个序列不会成功的

26:34 So give them the ability to look back, inspect
所以让他们有能力回过头去检查

26:37 or try to find, try to basically sample around it.
或者试着找到,尝试基本上围绕它进行抽样。

sample
样本 样本
神经网络基础

26:41 Here's one technique also.
还有一个技术

26:42 You could, it turns out that actually LLMs,
事实证明,实际上LLMS,

26:45 like they know when they've screwed up.
喜欢知道当他们搞砸了。

26:47 So as an example, say you asked the model
比如说,你问了模型

26:50 to generate a poem that does not rhyme
来产生一首不押韵的诗

26:53 and it might give you a poem, but it actually rhymes.
可能给你一首诗,但实际上它押韵。

26:56 But it turns out that especially for the bigger models
但结果发现,特别是对于大模型

26:58 like GPT-4, you can just ask it,
就像GPT -4 你可以直接问它

gpt
生成式预训练Transformer模型 生成式预训练Transformer模型
神经网络基础

26:59 did you meet the assignment?
你见过任务吗?

27:01 And actually GPT-4 knows very well
并且实际上GPT-4 很清楚

> **gpt**
> 生成式预训练Transformer模型 生成式预训练Transformer模型
> 神经网络基础

27:03 that it did not meet the assignment.
并且不符合任务。

27:05 It just kind of got unlucky in its sampling.
采样时有点不走运

27:07 And so it will tell you,
它将告诉你,

27:08 no, I didn't actually meet the assignment.
不,我其实没有遇到任务。

27:09 Here's, let me try again.
让我再试一次

27:11 But without you prompting it, it doesn't even,
但是没有你推动它, 它甚至没有,

27:13 like it doesn't know to revisit and so on.
喜欢它不知道 重新审视等等。

27:18 So you have to make up for that in your prompts.
所以你得用你的提示弥补

27:20 You have to get it to check.
你得去检查一下

27:22 If you don't ask it to check,
如果你不要求检查

27:23 it's not gonna check by itself,
自己检查也不行

27:24 it's just a token simulator.
这只是一个象征性的模拟器

27:29 I think more generally a lot of these techniques
我想一般而言,这些技巧

27:30 fall into the bucket of what I would say,
掉进我所说的话的桶里

27:32 recreating our system too.
重建我们的系统

27:34 So you might be familiar with the system
所以,你可能熟悉这个系统

27:36 one, system two, thinking for humans.
一,系统二,人类的思维

27:38 System one is a fast automatic process
系统1是一个快速自动程序

27:40 and I think kind of corresponds to like an LLN
我觉得有点像LLN

27:42 just sampling tokens.
仅仅采样象征物。

27:43 And system two is the slower deliberate planning
第二系统是慢于 精心策划的计划

27:47 sort of part of your brain.
某种大脑的一部分。

27:49 And so this is a paper actually from just last week
所以这是一份报纸 实际上来自上周

27:51 because this space is pretty quickly evolving.
因为这空间正在迅速演变

27:53 It's called Tree of Thought.
这叫《思想树》

27:55 And in Tree of Thought,
和在有理智的树上,

27:57 the authors of this paper proposed
本文件的作者建议

27:58 maintaining multiple completions for any given prompt
保持给定时间的多个补全

28:02 and then they are also scoring them along the way
然后,他们在大地上得分,

28:05 and keeping the ones that are going well, if that makes sense.
并保持运行良好,如果这合理的话。

28:08 And so a lot of people are like really playing around
所以很多人就像玩游戏

28:10 with kind of prompt engineering
具有某种即时工程

28:13 to basically bring back some of these abilities
基本上可以恢复一些这些能力

28:16 that we sort of have in our brain for LLMs.
我们大脑里有某种 用于LLMS的

28:19 Now, one thing I would like to note here
现在,我要在此指出一件事

28:21 is that this is not just a prompt.
这不仅仅是一个迅速的时机。

28:22 This is actually prompts that are together
这其实是一连在一起的提示

28:25 used with some Python glue code
与一些 Python 胶水代码使用

28:27 because you actually have to maintain multiple prompts
因为实际上您必须保持多个提示

28:30 and you also have to do some tree search algorithm here
并且你还必须在这里做一些树搜索算法

algorithm ['ælgəriðm]
算法 An effective procedure: a finite sequence of unambiguous steps that, when carried out on a given input, produces the intended output and terminates. The word derives from
神经网络基础

28:32 to like figure out which prompt to expand, et cetera.
喜欢找出什么能迅速扩张,等等。

28:35 So it's a symbiosis of Python glue code
所以这是皮顿胶水代码的共生关系

28:37 and individual prompts that are called in a Y loop
Y 循环中调用的个人提示和单个提示

28:40 or in a bigger algorithm.
或更大的算法。

algorithm ['ælgəriðm]
算法 An effective procedure: a finite sequence of unambiguous steps that, when carried out on a given input, produces the intended output and terminates. The word derives from
神经网络基础

训练数据 · 大语言模型 28:42

28:42 I also think that there's a really cool parallel
我也觉得这有种很酷的平行

28:44 here to AlphaGo.
这里是阿尔法戈

28:45 AlphaGo has a policy for placing the next stone
AlphaGo有下石板的保单

28:47 when it plays Go.
当它播放时,它会弹奏。

28:49 And this policy was trained originally
最初是经过训练的

28:50 by imitating humans.
模仿人类

28:52 But in addition to this policy,
但是除了这项政策之外

28:54 it also does multi-carload tree search
也进行多载量树搜索

28:56 and basically it will play out a number of possibilities
基本上它会发挥一些可能性

28:58 in its head and evaluate all of them
头部中,你应当加以统统地加以评估,并加以估价;

29:00 and only keep the ones that work well.
并且只保持那些工作良好的人。

29:01 And so I think this is kind of an equivalent of AlphaGo
所以我觉得这相当于阿尔法戈

29:05 but for text, if that makes sense.
但要是文本,如果这合理的话

29:09 So just like tree of thought, I think more generally,
所以就像一棵思想树,我想更一般地说

29:11 people are starting to like really explore
人开始喜欢探索

29:13 more general techniques of not just a simple question
不仅一个简单的问题,而且使用更一般的技巧

29:16 answer prompts, but something that looks a lot more
答案提示, 但看起来更像

29:18 like Python glue code stringing together many prompts.
就像皮顿胶水代码 串线在一起很多提示。

29:22 So on the right, I have an example
所以右边,我举个例子

29:23 from this paper called React where they structure
本文中的"反应"是它们结构的

29:26 the answer to a prompt as a sequence of thought, action,
答案迅速,作为一个一系列的想法,行动,

29:31 observation, thought, action, observation.
观察、思考、行动、观察。

29:34 And it's a full rollout, kind of a thinking process
这是一个全面展开的, 一种思考过程

29:36 to answer the query.
回答查询。

29:38 And in these actions, the model is also allowed to tool use.
在这些行动中,模型也允许工具使用。

29:42 On the left, I have an example of ROGPT.
左边,我有一个ROGPT的例子

29:45 And now ROGPT, by the way, is a project
顺便说一下,ROGPT是一个项目

29:48 that I think got a lot of hype recently.
我觉得最近有很多的 杂音。

29:52 But I still find it kind of inspirationally interesting.
但我还是觉得这有点灵感的有趣

29:56 It's a project that allows an LLM to keep task list
这是一个允许LLM 保持任务列表的项目

llm
大语言模型 大语言模型
神经网络基础

29:59 and continue to recursively break down tasks.
并不断重复拆分任务。

30:01 And I don't think this currently works very well,
我觉得目前这工作不怎么顺利

30:03 and I would not advise people to use it
我不会建议人们使用它

30:05 in practical applications.
实际应用。

30:07 I just think it's something to generally take inspiration
我只是觉得这一般是灵感的灵感

30:09 from in terms of where this is going, I think, over time.
我想,从未来到将来的 时间上看

30:13 So that's kind of like giving our model system to thinking.
这就像让我们的模型系统思考。

30:16 The next thing that I find kind of interesting
下一件我发现很有趣的事

30:17 is this following sort of, I would say,
这是一种以下的,我想说,

30:20 almost psychological quirk of LLMs
几乎在心理上紧张的长效微粒

30:23 is that LLMs don't want to succeed.
也就是LLMS不想成功

30:26 They want to imitate.
他们想模仿

30:28 You want to succeed, and you should ask for it.
你想成功,你应该要求它。

30:31 So what I mean by that is when transformers are trained,
所以我的意思是 当变压器经过训练时

30:35 they have training sets.
他们有训练装备

training ['treiniŋ]
训练 确定构成模型的理想参数 的过程。
神经网络基础

30:36 And there can be an entire spectrum of performance qualities
并且可以有一系列的性能品质

30:40 in their training data.
培训数据。

training ['treiniŋ]
训练 确定构成模型的理想参数 的过程。
神经网络基础

30:41 So for example, there could be some kind of a prompt
比如说,可以有某种迅速的

分布 · 概率 30:43

30:43 for some physics question or something like that.
来回答物理问题之类的

30:45 And there could be a student solution
学生可以找到一个解决办法

30:46 that is completely wrong,
这确是完全错误的。

30:48 but there can also be an expert answer
但也可能有专家的回答

30:49 that is extremely right.
这是极正确的。

30:51 And transformers can't tell the difference
变压器分辨不出区别

30:53 between, like, look, I mean, they know about
之间,就像,你看, 我的意思是,他们知道

30:55 low-quality solutions and high-quality solutions,
低质量解决办法和高质量解决办法,

30:58 but by default, they want to imitate all of it
但他们想仿造它,

31:00 because they're just trained on language modeling.
因为他们只是接受语言模型学的培训

31:02 And so at test time,
所以在测试时

31:03 you actually have to ask for a good performance.
您实际上必须要求良好的表现。

31:06 So in this example in this paper,
因此,在本文的这个例子中,

31:07 it's they tried various prompts
因为他们尝试了各种提示

31:10 and let's think step-by-step was very powerful
让我们想想,一步一步走是 强大的力量

31:13 because it's sort of like spread out
因为它有点像分散

31:14 the reasoning over many tokens,
有许多迹象,

31:15 but what worked even better is,
但更好的是,

31:17 let's work this out in a step-by-step way
让我们逐步解决这件事

31:19 to be sure we have the right answer.
确定我们有正确的答案

31:21 And so it's kind of like conditioning
所以它有点像调制

31:22 and getting a right answer,
并获得一个正确的答案,

31:23 and this actually makes the transformer work better
这实际上使变压器的工作效果更好

transformer [træns'fɔ:mə]
Transformer 中文为"变换器"，但不够达雅，日常还是称英文 Transformer 中文为"变换器"，但不够达雅，日常还是称英文
神经网络基础

31:26 because the transformer doesn't have to now hedge
因为变压器现在不用对冲了

transformer [træns'fɔ:mə]
Transformer 中文为"变换器"，但不够达雅，日常还是称英文 Transformer 中文为"变换器"，但不够达雅，日常还是称英文
神经网络基础

31:28 its probability mass on low-quality solutions
低质量解决方案的概率质量

probability
概率 概率
数学与微积分

31:31 as ridiculous as that sounds.
听起来很可笑

31:32 And so basically don't feel free to ask for a strong solution.
所以基本上,我们不觉得可以要求一个强有力的解决方案。

31:37 Say something like, you are a leading expert on this topic.
说点什么吧 你是这个话题的顶尖专家

31:39 Pretend you have IQ 120, et cetera.
假装你拥有120智商,等等。

31:42 But don't try to ask for too much IQ
但不要要求太多智商

31:43 because if you ask for IQ of, like, 400,
因为如果你要求智商400,

31:46 you might be out of data distribution.
您可能没有数据发布。

distribution
分布 分布
数学与微积分

31:48 Or even worse, you could be in data distribution
更糟糕的是,你可能参与数据传播

distribution
分布 分布
数学与微积分

31:51 for some like sci-fi stuff
某些像科幻的东西

31:52 and it will start to like take on some sci-fi,
并且开始喜欢 一些科学的,

31:54 like role-playing or something like that.
扮演角色之类的

31:57 So you have to find like the right amount of IQ,
所以你得找到 正确的智商

31:59 I think it's got some U-shaped curve there.
我想那里有U形曲线

32:03 Next up, as we saw, when we are trying to solve problems,
接下来,正如我们看到的, 当我们试图解决问题时,

32:07 we know what we are good at and what we're not good at
我们知道我们擅长什么,我们不擅长什么

32:09 and we lean on tools computationally.
我们用工具来计算。

32:12 You want to do the same potentially with your LLMs.
你想做同样的可能性 与你的LLMS。

32:15 So in particular, we may want to give them
特别是,我们或许想给他们

32:19 calculators, code interpreters, and so on,
计算器、密码翻译等等

32:22 the ability to do search.
能够进行搜索。

32:23 And there's a lot of techniques for doing that.
这样做有很多技巧

32:27 One thing to keep in mind again is that these transformers
需要记住的一件事 就是这些变压器

32:29 by default may not know what they don't know.
默认情况下可能不知道他们不知道什么

32:33 So you may even want to tell the transformer in a prompt,
所以你可能甚至想 告诉变压器 在迅速,

transformer [træns'fɔ:mə]
Transformer 中文为“变换器”，但不够达雅，日常还是称英文 Transformer 中文为“变换器”，但不够达雅，日常还是称英文
神经网络基础

32:35 you are not very good at mental arithmetic.
您在心理算术上并不擅长。

32:37 Whenever you need to do very large number addition,
每当您需要大量添加时,

32:40 multiplication or whatever, instead use this calculator.
乘法或其它什么的, 代之以使用此计算器 。

32:42 Here's how you use the calculator.
这就是你如何使用计算器的方法

32:43 Use this token, combination, et cetera, et cetera.
使用此符号、 组合等等等 。

32:46 So you have to actually like spell it out
所以,你必须真正喜欢 拼出来

32:47 because the model by default doesn't know
因为模型默认时不知道

Transformer 中文为“变换器”，但 32:49

32:49 what it's good at or not good at necessarily,
是否善于或不善于必有

32:51 just like you and I might be.
就像你和我一样

32:55 Next up, I think something that is very interesting
接下来,我想有些事很有趣

32:57 is we went from a world that was retrieval only
我们从一个只回收的世界

33:01 all the way the pendulum has swung to the other extreme
钟摆一直摇晃到另一个极端

33:04 where it's memory only in LLMs.
里面只有LLMS的记忆

33:06 But actually there's this entire space in between
但实际上,在

33:08 of these retrieval augmented models
增强型号

33:10 and this works extremely well in practice.
在实践中,这非常有效。

33:13 As I mentioned, the context window of a transformer
正如我提到,变压器的上下文窗口

transformer [træns'fɔ:mə]
Transformer 中文为"变换器"，但不够达雅，日常还是称英文 Transformer 中文为"变换器"，但不够达雅，日常还是称英文
神经网络基础

33:15 is it's working memory.
这是一种工作记忆

33:17 If you can load the working memory
如果您可以装入工作内存

33:18 with any information that is relevant to the task,
附上与任务有关的任何资料,

33:21 the model will work extremely well
模型将非常有效

33:23 because it can immediately access all that memory.
因为它能立即访问所有记忆

33:26 And so I think a lot of people are really interested
所以我觉得很多人都非常感兴趣

33:28 in basically retrieval augmented generation.
基本上检索 增强生成。

33:33 And on the bottom I have like an example of Lama index
底部我有个拉马指数的例子

33:35 which is one sort of data connector
是一个数据连接器

33:37 to lots of different types of data
或多种不同类型数据

33:38 and you can index all of that data
,然后您就可以将所有的数据索引

33:42 and you can make it accessible to LLMs.
并且可以让LLMS进入

33:44 And the emerging recipe there is you take relevant documents,
正在形成的食谱是 你拿了相关的文件

33:47 you split them up into chunks, you embed all of them
你把他们分成了块块, 把它们都嵌入

33:50 and you basically get embedding vectors
基本上,你得到嵌入矢量

33:52 that represent that data.
表示该数据。

33:53 You store that in the vector store
您将存储在矢量仓库

33:55 and then at test time you make some kind of a query
然后在测试时间,您会做某种查询

33:57 to your vector store and you fetch chunks
转到向量存储处,然后获取块

33:59 that might be relevant to your task
可能与任务有关

34:01 and you stuff them into the prompt and then you generate.
然后,你把他们投入到急迫的状态中,然后,你创造出来。

34:04 So this can work quite well in practice.
因此,这在实践中可以相当顺利。

34:06 So this is I think similar to when you and I solve problems,
所以我的想法和我们解决问题时的相似

34:09 you can do everything from your memory
记忆中的一切你都可以做

34:11 and transformers have very large and extensive memory
变压器的内存量巨大且广泛

34:13 but also it really helps to reference
也确实有助于引用

34:16 some primary documents.
一些主要文件。

34:17 So whenever you find yourself going back to a textbook
所以当你发现自己回到教科书

34:20 to find something or whenever you find yourself going back
找到某事或当你发现自己回去时

34:22 to documentation of a library to look something up,
文档到一个图书馆的文档 查找的东西上,

34:24 the transformers definitely wanna do that too.
变压器也肯定想这么做

34:27 You have some memory over how some documentation
您对文件的多少记忆力有些

34:31 of a library works but it's much better to look it up.
图书馆的工作,但它是更好的 查起来。

34:33 So the same applies here.
情况也是如此。

34:37 Next I wanted to briefly talk about constraint prompting.
接下来我想简要谈谈促使限制的因素。

34:39 I also find this very interesting.
我也觉得这很有趣

人类反馈的强化学习 · 微调 34:42

34:42 This is basically techniques for
这基本上是用来

34:46 forcing a certain template in the outputs of LLMs.
强制在LLMM产出中设置某个模板。

34:50 So guidance is one example from Microsoft actually.
所以指导其实是微软的一个实例。

34:53 And here we are enforcing that the output from the LLM
我们在这里执行 有限责任协议的产出

llm
大语言模型 大语言模型
神经网络基础

34:56 will be JSON. 将JSON。
将杰森。

34:57 And this will actually guarantee that the output
这将保证输出

35:00 will take on this form because they go in
将采用这个表格,因为他们进去了

35:02 and they mess with the probabilities
他们将用计谋,他们以计谋,

35:03 of all the different tokens that come out of the
不同物的象征 从

35:05 transformer and they clamp those tokens.
变压器和他们固定这些标志。

transformer [træns'fɔ:mə]
Transformer 中文为"变换器"，但不够达雅，日常还是称英文 Transformer 中文为"变换器"，但不够达雅，日常还是称英文
神经网络基础

35:07 And then the transformer is only filling in the blanks
然后变压器只填满空的

transformer [træns'fɔ:mə]
Transformer 中文为"变换器"，但不够达雅，日常还是称英文 Transformer 中文为"变换器"，但不够达雅，日常还是称英文
神经网络基础

35:09 here and then you can enforce additional restrictions
在此, 然后您可以执行额外的限制

35:11 on what could go into those blanks.
如何进入这些空白。

35:13 So this might be really helpful and I think
所以这也许真的有用 我认为

35:15 this kind of constraint sampling is also extremely interesting.
这种限制抽样也非常有趣。

35:20 I also wanted to say a few words about fine tuning .
我还想说几句话,讲一下微调。

fine tuning
微调 微调
神经网络基础

35:22 It is the case that you can get really far
情况是,你可以走得非常远

35:24 with prompt engineering but it's also possible
工程工程的迅速,但也是可能的

35:27 to think about fine tuning your models.
考虑如何调整你的模型

fine tuning
微调 微调
神经网络基础

35:29 Now fine tuning models means that you are actually
现在微微调模型意味着你实际上

fine tuning
微调 微调
神经网络基础

35:31 going to change the weights of the model.
将改变模型的重量。

weights
权重 权重
神经网络基础

35:34 It is becoming a lot more accessible to do this
这样做的便利性正在变得更大

35:36 in practice and that's because of a number of techniques
是因为一些技巧

35:39 that have been developed and have libraries for
所开发的图书馆和拥有图书馆

35:42 very recently.
近期才开始

35:43 So for example, parameter efficient fine tuning techniques
例如,参数高效微调技术

fine tuning
微调 微调
神经网络基础

35:45 like Laura, make sure that you're only training
像劳拉一样,确保你只是训练

training ['treiniŋ]
训练 确定构成模型的理想参数 的过程。
神经网络基础

35:48 small sparse pieces of your model.
微小的零散碎片 你的模型。

35:51 So most of the model is kept clamped at the base model
所以大部分模型都一直被基模压住

35:54 and some pieces of it are allowed to change
一部分,可以变更

35:56 and this still works pretty well empirically
并且这个仍然工作 相当良好的经验

35:58 and makes it much cheaper to sort of tune
并且让调子更便宜

36:00 only small pieces of your model.
只不过你模型的小块

36:02 It also means that because most of your model is clamped,
也意味着因为你的型号大部分都被夹住了

36:05 you can use very low precision inference
您可以使用非常低精密的推断

36:08 for computing those parts because they are not
计算这些部件,因为它们不是

36:10 going to be updated by gradient descent
将按梯度下降情况更新

36:11 and so that makes everything a lot more efficient as well.
这样也使一切变得效率更高

36:14 And in addition, we have a number of open sourced
此外,我们还有一系列开放源

36:16 high quality based models.
高质量的模型。

36:17 Currently, as I mentioned, I think Lama is quite nice
正如我提到,目前拉马人很好

36:20 although it is not commercially licensed,
虽然它没有商业许可证,

36:22 I believe right now.
我现在就相信了

36:24 Something to keep in mind is that basically
需要记住的是,基本上

36:26 fine tuning is a lot more technically involved.
微调在技术上要多得多。

fine tuning
微调 微调
神经网络基础

36:29 It requires a lot more, I think technical expertise
还需要很多,我想需要技术专长

36:31 to do right.
做对了

36:32 It requires human data contractors for data sets
要求数据集需要人数据承包商

36:35 and or synthetic data pipelines
或合成数据管道

36:37 that can be pretty complicated.
可能相当复杂

36:38 This will definitely slow down your iteration cycle
这肯定会减慢你的循环周期

36:40 by a lot and I would say on a high level,
有很多事,我会说 在高层次上,

36:43 SFD is achievable because it is just
SAFD是可以实现的,因为它是公正的

36:46 you're continuing the language modeling task.
您正在继续语言模型的工作。

36:48 It's relatively straightforward but RLHF, I would say,
相对简单,但RLHF,我想说

> **rlhf**
> 人类反馈的强化学习 人类反馈的强化学习
> 神经网络基础

36:51 is very much research territory
研究领域非常多

36:53 and is even much harder to get to work.
工作更难

36:55 And so I would probably not advise that someone just tries
所以我可能不建议有人 仅仅尝试

36:58 to roll their own RLHF implementation.
以滚动实施自己的RLHF。

> **rlhf**
> 人类反馈的强化学习 人类反馈的强化学习
> 神经网络基础

37:00 These things are pretty unstable, very difficult to train,
这些东西很不稳定,很难训练

37:03 not something that is I think very beginner friendly
而不是我认为非常友好的

37:05 right now and is also potentially likely also
并且也有可能

大语言模型 · 人类反馈的强化学习 37:08

37:08 to change pretty rapidly still.
变化很快

37:11 So I think these are my sort of default recommendations
所以我认为这是我的 违约建议

37:14 right now.
现在

37:15 I would break up your task into two major parts.
我会把你们的任务分成两大部分

37:17 Number one, achieve your top performance
第一,完成你最优秀的成绩

37:20 and number two, optimize your performance in that order.
第二,按照这个顺序优化你的性能

37:24 Number one, the best performance will currently
第一,目前最佳表现

37:25 come from GPD4 model.
取自GPD4模型。

37:27 It is the most capable model by far.
它是迄今为止最有能力的模式。

37:29 Use prompts that are very detailed.
使用非常详细的提示。

37:31 They have lots of task contents,
他们有许多任务,

37:33 relevant information and instructions.
有关资料和说明。

37:36 Think along the lines of what would you tell
想想你会说什么

37:37 a task contractor if they can't email you back
任务承包商如果无法回信给你

37:40 but then also keep in mind that a task contractor
但同时要铭记,任务承包人

37:43 is a human and they have inner monologue
人与人之间,有独白

37:44 and they're very clever, et cetera.
他们非常聪明,等等。

37:46 LLMs do not possess those qualities.
女优不具有这些品质。

37:48 So make sure to think through the psychology of the LLM
所以一定要通过LLM的心理学思考

llm
大语言模型 大语言模型
神经网络基础

37:52 almost and cater prompts to that.
几乎满足了这个要求

37:55 Retrieve and add any relevant context
检索并添加任何相关上下文

37:57 and information to these prompts.
以及这些消息的真伪。

38:00 Basically refer to a lot of the prompt engineering techniques.
基本上是指许多快速工程技术。

38:03 Some of them I've highlighted in the slides above
上面的幻灯片中突出提到过其中的一些

38:05 but also this is a very large space
但也是一个很大空间

38:07 and I would just advise you to look for
我建议你找找

38:10 prompt engineering techniques online.
在线快速工程技术。

38:11 There's a lot to cover there.
那里有很多东西要盖

38:13 Experiment with few shot examples.
实验用几个实例。

38:15 What this refers to is you don't just wanna tell,
这指的是 你不只是想说

38:17 you want to show whenever it's possible.
你想在可能的时候显示

38:19 So give it examples of everything
故你当以一切事物为例,

38:22 that helps it really understand what you mean if you can.
帮助它真正理解你的意思 如果你可以的话

38:25 Experiment with tools and plugins to offload a task
使用工具及插件进行实验, 以卸载任务

38:28 that are difficult for LLMs natively
当地LLM女士难以

38:31 and then think about not just a single prompt
然后,你应当想一想,不要只做一个计时的、

38:32 and answer, think about potential chains and reflection
并回答,思考潜在的链和反思

38:35 and how you glue them together
你们怎样使他们互相粘合,

38:37 and how you could potentially make multiple samples and so on.
以及你如何能做出多个样本等等

38:40 Finally, if you think you've squeezed out prompt engineering
如果你最后认为你已经挤出工程

38:43 which I think you should stick with for a while,
我认为你应该坚持一段时间,

38:45 look at some potentially fine tuning a model
审视一些可能微调模型的

fine tuning
微调 微调
神经网络基础

38:50 to your application but expect this to be
申请时,您必须希望

38:52 a lot more slower and involved.
慢得多,多参与。

38:54 And then there's an expert fragile research zone here
然后这里就有一个 专家脆弱的研究区

38:56 and I would say that is RLHF
我要说的是,RLHF

rlhf
人类反馈的强化学习 人类反馈的强化学习
神经网络基础

38:58 which currently does work a bit better than SFD
其目前的工作效果比SFD好

39:01 if you can get it to work.
如果你能让它工作的话

39:02 But again, this is pretty involved I would say.
但再说一遍,我还是要说这很牵扯

39:05 And to optimize your costs, try to explore
并尽量优化成本,尝试探索

39:07 lower capacity models or shorter prompts and so on.
低能力模型或短速提示等等。

39:13 I also wanted to say a few words about the use cases
我还想就使用案例讲几句话

39:15 in which I think LLMs are currently well suited for.
我认为,LLMS目前很适合。

39:18 So in particular, note that there's a large number
特别是,请注意,有一大堆

39:20 of limitations to LLMs today.
限制今天的LLMs。

39:22 And so I would keep that definitely in mind
所以我会记住

39:25 for all your applications.
申请的所有申请。

39:26 Models, and this by the way could be an entire talk
模特儿,顺便说一下,这可以是一个整场谈话

39:28 so I don't have time to cover it in full detail.
所以我没时间详细报道

39:30 Models may be biased, they may fabricate,
模式可能带有偏见,它们可能编造,

39:32 hallucinate information, they may have reasoning errors,
幻觉信息,他们可能有推理错误

39:35 they may struggle in entire classes of applications.
可能在整个申请类别中挣扎

39:38 They have knowledge cutoffs so they might not know
他们有知识,故不能知道;

39:41 any information above say September 2021.
上面的任何信息,如2021年9月。

39:44 They are susceptible to a large range of attacks
他们容易受到各种攻击

39:46 which are sort of like coming out on Twitter daily
像是每日在推特上发布

39:48 including prompt injection, jailbreak attacks,
包括及时注射、越狱袭击,

39:51 data poisoning attacks and so on.
数据中毒袭击等等。

39:53 So my recommendation right now is use LLMs
所以我现在的建议是使用LLMS

39:55 in low stakes applications,
低利害关系申请,

39:57 combine them with always with human oversight,
与人的监督相结合

40:00 use them as a source of inspiration and suggestions
利用它们作为灵感和提出建议的来源

40:03 and think co-pilots instead of completely autonomous agents
思考副驾驶员而不是完全自主的代理

40:05 that are just like performing a task somewhere.
就像是在某处执行任务

生成式预训练Transformer模型 40:08

40:08 It's just not clear that the models are there right now.
只是现在还不清楚模型是否在现场

40:12 So I wanted to close by saying that GPT -4
所以我想最后说一下 GPT-4

gpt
生成式预训练Transformer模型 生成式预训练Transformer模型
神经网络基础

40:14 is an amazing artifact, I'm very thankful that it exists
是一个惊人的艺术品, 我很感激它的存在

40:17 and it's beautiful, it has a ton of knowledge
美丽,知识丰富

40:19 across so many areas, it can do math, code and so on.
跨越如此多的领域, 它可以做数学、代码等等。

40:23 And in addition there's this thriving ecosystem
此外,还有这个繁荣的生态系统

40:25 of everything else that is being built
其它一切正在建设的

40:27 and incorporated into the ecosystem.
并融入生态系统。

40:29 Some of these things I've talked about.
有些事情我说过

40:32 And all of this power is accessible at your fingertips.
所有的这些力量都可以从你的指尖上获得

40:35 So here's everything that's needed in terms of code
这就是代码方面需要的一切

40:38 to ask GPT -4 a question, to prompt it and get a response.
向GPT-4提出一个问题, 并迅速得到答复。

> **gpt**
> 生成式预训练Transformer模型 生成式预训练Transformer模型
> 神经网络基础

40:42 In this case I said, can you say something
我说,这个案子 你能说点什么

40:45 to inspire the audience of Microsoft Build 2023?
激励微软2023年建筑的观众?

40:48 And I just punched this into Python and verbatim,
我把这个打进Python和逐字记录

40:51 GPT -4 said the following. GPT -4说:
GPT-4说:

> **gpt**
> 生成式预训练Transformer模型 生成式预训练Transformer模型
> 神经网络基础

40:54 And by the way I did not know
顺便说一句,我不知道

40:55 that they used this trick in the keynote.
并且他们用这个把戏 在主旨。

40:57 So I thought I was being clever but,
所以我以为我很聪明,但是

41:00 but it is really good at this.
但它确实擅长于此。

41:02 It says, ladies and gentlemen, innovators
上面说,女士们先生们,创新者

41:04 and trailblazers of Microsoft Build 2023,
以及微软2023年建筑的散星。

41:07 welcome to the gathering of brilliant minds like no other.
欢迎来到智者聚会 和其他人一样

41:10 You are the architects of the future,
你们是未来的建筑师

41:11 the visionaries molding the digital realm
数字领域塑造的愿景

41:13 in which humanity thrives,
人类在其中繁衍,

41:15 embrace the limitless possibilities of technologies
技术的无限可能性

41:18 and let your ideas soar as high as your imagination.
让你的想法飞快地飞扬,就像你的想象力一样

41:20 Together let's create a more connected, remarkable
让我们一起创造更连接, 更显著的

41:23 and inclusive world for generations to come.
以及子孙后代的包容性世界。

41:25 Get ready to unleash your creativity,
准备释放你的创造力

41:27 canvass the unknown and turn dreams into reality.
将梦想变成现实

41:30 Your journey begins today.
你的旅程从今天开始

41:33 Thank you.
谢谢

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